

Adapter Merging on JEPA Backbones: Interference Lives in the Head, and a Rate–Distortion–Optimal Encoder Recovers It

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Abstract

Weight-space merging of independently trained LoRA adapters is now standard practice for CLIP- and LLM-family encoders, but has never been studied on Joint-Embedding Predictive Architectures (JEPAs), despite their growing role as general visual backbones. We present the first study of PEFT adapter merging on JEPA-family encoders (V-JEPA 2, plus MAE, DINOv2, and LeJEPA backbones for contrast), evaluating five mergers under two protocols. Our central empirical finding is a *localization of interference*: under a re-fit linear probe (P1), merged encoders retain near-perfect accuracy (mean retention 1.06), but under reuse of the original task heads (P2), averaging and TIES-style mergers lose 15–24 points (mean retention 0.835). The P1–P2 gap is statistically robust (paired Wilcoxon $p = 6.8 \times 10^{-7}$, 5 backbones, 50 merges) and shows that at $k=2$, merging on JEPA backbones does not destroy linearly-decodable features—it breaks head-compatibility. Whereas re-fitting a probe on supervised-ViT merges only *partially* restores accuracy [Sharma et al., 2024], here recovery is essentially complete, localizing interference almost entirely to the head. Second, a rate–distortion–optimal merging encoder (RD-encoder) is the strongest head-preserving merger: it has the smallest P1–P2 gap of any method (0.04 vs 0.19–0.43), is best on P2 for four of five backbones, and beats task arithmetic on P2 overall by +0.19 (95% CI [+0.09, +0.28]). The exception is DINOv2, which already merges unusually well (its averaging/TIES baselines are the highest of any backbone), leaving little headroom and giving TIES a slight edge there. We release the metric library, adapters, and a SIGReg isotropy-dial pretraining harness. A causal study of representation isotropy and composability is motivated by these results and is ongoing; in this pilot the isotropy dial did not yield a controllable downstream-isotropy gradient (Section 6), which we report honestly as a negative methodological result.

1 Introduction

Model merging combines independently fine-tuned adapters into one multi-task model with no joint training. The literature is CLIP/LLM-centric; JEPA-family encoders—now competitive general visual backbones—have not been studied under merging at all. We ask the basic questions first: *does merging work on JEPA backbones, where does interference appear, and which merger is most robust?* We find that the answer depends sharply on the evaluation protocol, and that a rate–distortion–optimal encoder is markedly more robust than the standard averaging and trimming mergers.

Contributions. (1) The first study of PEFT adapter merging on JEPA-family encoders. (2) A *sharpened* interference-localization result. Prior work on supervised-ViT merging used the same probe-refit instrument and found that re-fitting only *partially* recovers merged-model accuracy [Sharma et al., 2024]; we show that

on JEPA-family SSL backbones at $k=2$ the re-fit probe recovers to ceiling (P1 mean 1.04) while original-head accuracy (P2) drops 15–24 points—interference is localized *almost entirely* to head-compatibility, not partially. The conceptual frame (probe re-fitting to localize interference) is not ours; the completeness of the recovery on these backbones is the empirical contribution. (3) Our principal methodological contribution: a rate–distortion-optimal merging encoder (RD-encoder) that is the strongest head-preserving merger across SSL families and two scales (smallest P1–P2 gap; best on P2 for four of five backbones; it ties on the well-merging DINOv2). (4) An open release: metric library, adapters, and a SIGReg-based isotropy-dial pretraining harness with a documented collapse-avoidance fix.

2 Related work

Model merging and task vectors. Averaging fine-tuned weights (model soups Wortsman et al. [2022]) and task arithmetic on the fine-tuning deltas Ilharco et al. [2023] combine models without joint training. Interference between task vectors motivates sparsification and sign-election schemes: TIES Yadav et al. [2023] trims and elects signs, DARE Yu et al. [2024] randomly drops and rescales. A separate line treats the geometry of the merged update directly: Iso-C/Iso-CTS Marczak et al. [2025] flatten (isotropize) the *singular-value spectrum of the merged task-arithmetic matrix* at merge time. Our RD-encoder instead realizes the achievability construction of a rate–distortion view of merging Pathak and Garg [2026]. These spectrum/merge-time methods are orthogonal to our question—we ask where interference *appears* (features vs heads) on a family of backbones, and which existing merger is most robust there. All adaptation is via LoRA Hu et al. [2022].

Closest prior work. Two lines are directly adjacent. Kesim and Helli [2024] merge up to three LoRA adapters trained on six supervised vision tasks and compare against a frozen-backbone, fine-tuned-head baseline, reporting—as we do—that merging is achievable with some loss of performance; their setting is supervised ViT rather than JEPA-family SSL, and they do not separate feature- from head-level interference. Sharma et al. [2024] ask precisely *where* merging interference originates: they re-fit a linear probe on the merged model to test whether the penultimate-layer features remain compatible with the fixed expert classifier, and propose rescaling and shifting the output activations to compensate. This is the same probe-refit instrument we use, and the head-localization frame is theirs, not ours; on supervised ViT they find that probing only *partially* restores merged accuracy. Our contribution relative to them is empirical and sharper: on JEPA-family SSL backbones at $k=2$ the re-fit probe recovers *fully* (P1 mean 1.04), localizing interference almost entirely to the head, and we contribute a head-preserving merger (RD-encoder) that closes the P1–P2 gap without any re-fitting.

JEPA-family and self-supervised backbones. Joint-embedding predictive architectures predict masked targets in latent space Assran et al. [2023, 2025]; LeJEPA Balestrierio and LeCun [2025] gives a predictor-free objective (SIGReg) with a theoretical optimum at isotropic-Gaussian embeddings. We contrast against masked reconstruction He et al. [2022] and self-distillation Oquab et al. [2024]. To our knowledge no prior work studies adapter merging on JEPA-family encoders specifically.

Representation geometry. Anisotropy and dimensional collapse are well documented Gao et al. [2019]; we measure isotropy with effective rank Roy and Vetterli [2007] and IsoScore Rudman et al. [2022], and representation agreement with linear CKA Kornblith et al. [2019].

3 Setup

Backbones. V-JEPA 2 ViT-L (image input via single-frame replication, predictor skipped), MAE ViT-L/16, DINOv2 ViT-L/14, and two LeJEPA ViT-S/16 backbones pretrained in-house on ImageNet-100 differing only in SIGReg coefficient.

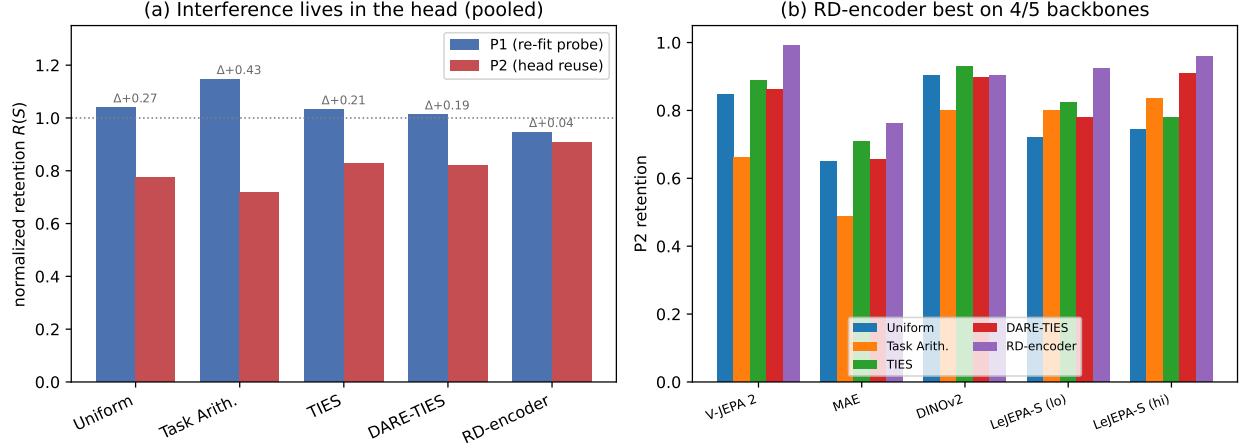


Figure 1: **(a)** Retention by merger under the two protocols, pooled over backbones and pairs. P1 (re-fit probe) sits at/above ceiling for every merger; P2 (head reuse) drops sharply for the averaging/trimming mergers. The gap Δ is the interference-localization statistic; RD-encoder’s gap is near zero. **(b)** P2 retention by backbone. RD-encoder (purple) is the best merger on four of five backbones; DINOv2, where all methods merge well, is the exception.

Adaptation. LoRA ($r=16, \alpha=32$, dropout 0.1) on the attention qkv and output-projection linears of every block; a task-private linear head is trained jointly and is never merged.

Tasks/merges. Pilot pairs eurosat–resisc45 (semantically close, both remote sensing) and mnist–dtd (far); five mergers (uniform, task arithmetic, TIES, DARE-TIES, RD-encoder) with hyperparameters frozen before evaluation (no per-result tuning).

Mergers. Uniform averages the task vectors; task arithmetic sums them with a tuned coefficient; TIES trims to the top-magnitude entries, elects a per-coordinate sign, and averages the agreeing entries; DARE-TIES adds random drop-and-rescale before TIES. RD-encoder forms a ridge-regularized, H -weighted centroid of the task deltas in the top- r right-singular “projector” geometry of each delta and realizes it as a full-rank weight update Pathak and Garg [2026]; merger hyperparameters (ridge $\lambda=0.05$) were frozen from an independent sweep.

Protocols and metric. P1 (primary): a fixed-budget logistic probe re-fit per task on the frozen merged encoder (immune to head-mismatch artifacts). P2 (secondary): reuse each task’s original fine-tuned head. We report normalized retention $R(S) = \frac{1}{|S|} \sum_{i \in S} \text{acc}_{\text{merged}}(i) / \text{acc}_{\text{solo}}(i)$ for a merge set S , with absolute accuracies in the release. The P1-vs-P2 contrast is the load-bearing comparison of this report.

4 Interference lives in the head

Table 1 and Figure 1(a) pool retention over backbones and pairs. Under P1 every merger is at or above ceiling (mean 1.04); under P2 the averaging/trimming mergers drop sharply. The paired P1-vs-P2 difference is significant (Wilcoxon $p = 6.8 \times 10^{-7}$, $n=50$). Interpretation: at $k=2$ on these backbones, the merged encoder still linearly encodes both tasks—a re-fit probe recovers them—so the interference that practitioners observe with frozen heads is a *head-compatibility* failure, not destruction of the underlying features. Where Sharma et al. [2024] report that probe re-fitting only partially restores merged supervised-ViT accuracy, here recovery is essentially complete (P1 mean 1.04), indicating that on these SSL backbones at low k the features are left almost intact. The practical reading is that on JEPA and related SSL backbones, a cheap per-task probe re-fit after merging recovers most of the apparent interference; the harder problem, which the next section addresses, is preserving the original heads.

Table 1: Retention by merger and protocol (C0, pooled over backbones+pairs). P1 re-fit probe vs P2 original-head reuse. The P1–P2 gap is the interference-localization statistic.

Merger	P1 (re-fit)	P2 (head reuse)	P1–P2 gap
uniform	1.039	0.773	0.265
task arithmetic	1.146	0.716	0.430
TIES	1.031	0.826	0.205
DARE-TIES	1.012	0.821	0.191
RD-encoder	0.945	0.907	0.038

Table 2: P2 (head-reuse) retention by backbone and merger (C0). RD-encoder is best on four of five backbones (bold); DINOv2, which merges unusually well under all methods, is the exception (TIES best).

Backbone	uniform	task-arith	TIES	DARE-TIES	RD-encoder
V-JEPA 2 ViT-L	0.848	0.660	0.889	0.860	0.990
MAE ViT-L	0.651	0.488	0.707	0.656	0.761
DINOv2 ViT-L	0.904	0.799	0.931	0.898	0.902
LeJEPA ViT-S (low)	0.719	0.801	0.824	0.779	0.922
LeJEPA ViT-S (high)	0.745	0.835	0.779	0.909	0.960

5 An RD-optimal encoder preserves heads

RD-encoder has the smallest P1–P2 gap of any merger (0.04 vs 0.19–0.43 for the baselines; Table 1): it preserves the original heads nearly as well as a re-fit probe preserves features. Across backbones (Table 2) it is best on P2 for four of five, with margin +0.10 (V-JEPA 2), +0.05 (MAE), +0.10/+0.05 (LeJEPA ViT-S) over the best averaging/TIES baseline, and +0.19 over task arithmetic overall (95% CI [+0.09, +0.28], $n=10$). DINOv2 is the exception: its baselines are the strongest of any backbone (TIES 0.93) and RD-encoder (0.90) essentially ties them (−0.03). Notably, RD-encoder’s margin over naive merging is largest exactly where naive merging interferes most (MAE, V-JEPA 2) and vanishes where it interferes least (DINOv2)—a useful property for a default merger.

Does task-vector geometry explain the gap? We logged pairwise cosine, per-coordinate sign-conflict rate, and top- r subspace overlap for every merge. Across the ten (backbone, pair) cells these do not cleanly predict baseline P2 interference: subspace overlap correlates with P2 retention in the *opposite* direction to the naive “more overlap $\rightarrow$ more interference” hypothesis (Spearman $\rho = 0.79$, but $n = 10$ and uncontrolled across heterogeneous backbones). We therefore do not claim a geometric mechanism here and release the per-merge geometry for scrutiny.

6 Representation isotropy: motivation and a negative pilot result

LeJEPA proves isotropic embeddings minimize worst-case downstream risk; a natural conjecture is that backbone isotropy also governs composability. We built a controlled dial: LeJEPA ViT-S backbones differing only in SIGReg coefficient λ . We report two honest pilot outcomes. (i) A naive predictor-free SIGReg objective *collapses* (embedding variance $\rightarrow 0$); the fix is BatchNorm on the pooled tokens before the loss, which we document and release. (ii) Even post-fix, our short-schedule pilot dial did not produce a controllable gradient in *downstream* (raw-feature) isotropy—the two backbones reached near-identical effective rank (85.8 vs 73.0)—so the causal isotropy–composability hypothesis could not be tested here. We attribute this to normalization decoupling SIGReg-space isotropy from raw-feature isotropy, and to schedule length; the full study addresses both. We present this as motivation and a documented negative result, not a confirmed claim.

7 Discussion

Two takeaways are robust across five backbones and two SSL scales. First, adapter merging on JEPA-family encoders does not destroy features at $k=2$; it breaks head-compatibility, and the P1-vs-P2 gap quantifies this. This reframes “merging interference” on these backbones as a head-alignment problem and suggests a cheap mitigation (probe re-fit) when re-fitting is permissible. Second, a rate-distortion-optimal encoder is the most head-preserving merger, and—usefully for a default choice—its advantage is largest exactly where naive merging hurts most and negligible where naive merging already works (DINOv2). The natural next questions are whether the localization persists at higher k (where feature-level interference should eventually appear) and across more task pairs, and whether an explicit head-alignment step closes the P1-P2 gap for the cheaper mergers.

8 Limitations

Single seed; pilot pairs ($k=2$) only; reduced fine-tuning and pretraining schedules (stated explicitly); ViT-L/ViT-S scale mismatch reported, never pooled. DINOv2 and MAE enter at ViT-L, the LeJEPA backbones at ViT-S. The isotropy dial is not yet a working causal instrument (Section 6), so we make no causal isotropy claim. The geometry mediators we logged did not yield a clean mechanism; we release them rather than over-interpret.

9 Reproducibility and release

We release, under a permissive license: the metric library (effective rank, IsoScore, sign-conflict, subspace-angle, CKA, retention) as an importable package; all LoRA adapters and merged checkpoints; the five merger implementations; and the LeJEPA SIGReg isotropy-dial pretraining harness including the Batch-Norm collapse fix (Section 6). Every number and figure in this report is regenerated by two scripts (`analyze_pivot.py`, `make_figures.py`) from the released per-merge JSON. Archived with a versioned DOI on Zenodo: doi:10.5281/zenodo.20711952 (concept DOI, resolves to the latest version); current version doi:10.5281/zenodo.20713190; source at <https://github.com/K144U/jepa-adapter-merging>.

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